Muscular System

Core Learning · Foundations of Human Anatomy & Physiology

Section 5: The Muscular System

5.1 What Is Muscle?

Muscle is a type of tissue that can create force and change length. There are three types of muscle in the body:

- **Skeletal muscle** — located beneath the layers of skin and fat; connects to tendons and bones; responsible for creating voluntary movement. Primary muscle type relevant to coaching.
- **Cardiac muscle** — found only in the heart; contracts and relaxes involuntarily to pump blood. Fatigue-resistant.
- **Smooth muscle** — forms the walls of blood vessels, glands, and organs; expands and contracts to move fluids. Involuntary.

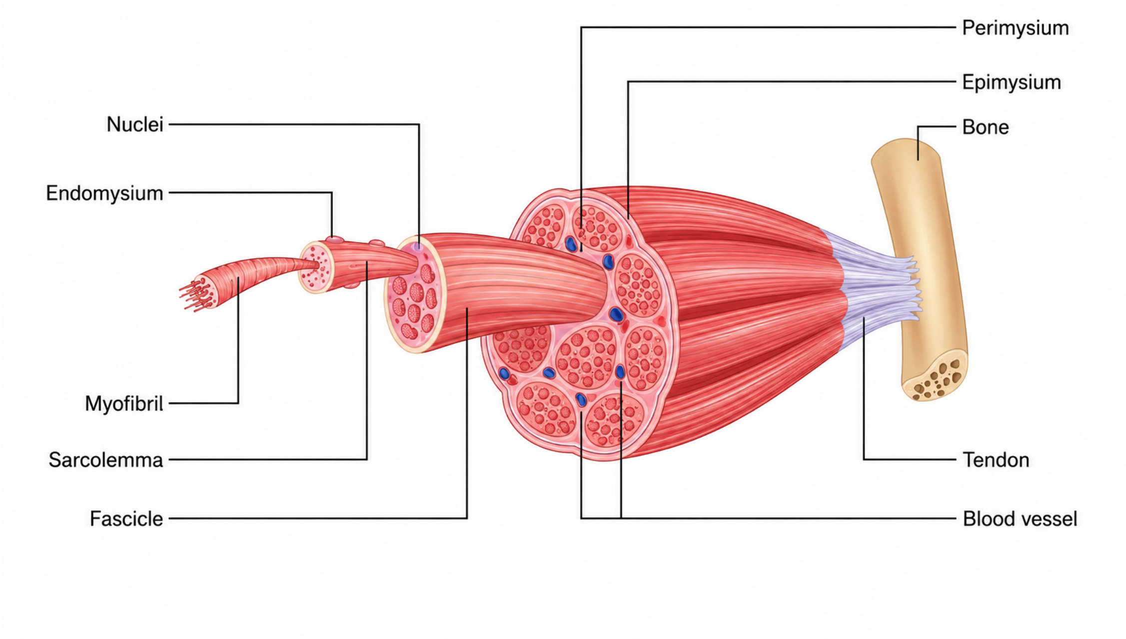
All muscle types share five unique properties:

1. **Irritability** — capable of receiving and responding to stimulation from nerves
2. **Contractibility** — once stimulated, the muscle actively shortens (contracts)
3. **Extensibility** — with the application of force, muscle can be stretched without damage
4. **Elasticity** — muscle returns to its resting shape and length after shortening or lengthening
5. **Adaptability** — muscle hypertrophies with increased work and atrophies if deprived of work. This underpins all training adaptation.

5.2 Skeletal Muscle Structure — From Fibre to Fascicle

Skeletal muscle is made up of thousands of muscle fibres that run the length of the muscle. This hierarchy explains why tendons and connective tissue respond more slowly to training than muscle tissue:

Structural Level	Description
Myofilaments	The thinnest level: actin (thin) and myosin (thick) protein filaments.
Myofibrils	The contractile units within each muscle fibre. Contain actin and myosin.
Sarcomeres	The smallest functional unit of skeletal muscle, found within myofibrils.
Muscle fibres	Individual muscle cells, wrapped in connective tissue (fascia).
Fascicles	Bundles of muscle fibres bound together by more fascia.
Tendon	Fascia thickens and coats the muscle, becoming the tendon — attaching muscle to bone.



From muscle fibre to whole muscle — the connective tissue hierarchy

5.3 Functions of Skeletal Muscle

1. **Force generation for movement** — every climbing move is created by coordinated muscle contraction.
2. **Force generation for breathing** — rib and abdominal muscles enable lung filling/emptying. Breathing rhythm is coachable.
3. **Force generation for postural support** — stabilises joints and maintains posture, minimising gravity's effects and injury risk.
4. **Heat production** — contracting muscles produce heat voluntarily (exercise) or involuntarily (shivering).

5.4 Muscle Origin and Insertion

- **Origin** — the relatively fixed end of the muscle, normally proximal. Provides the stable base the muscle pulls from.
- **Insertion** — the end that moves most, normally distal. Contraction pulls it toward the origin, creating joint movement.

Example: the biceps brachii has two origins on the scapula ('bi' = two), inserting on the radius. Contracting pulls the radius upward, flexing the elbow. The rotator cuff acts as fixators, stabilising the shoulder so the biceps can function effectively.

5.5 Muscle Roles During Movement

Muscle Role	Definition & Climbing Example
Agonist (Prime Mover)	Provides the major force to complete the movement. In a pull-up, the lats are the primary agonist — concentrically or eccentrically.
Antagonist	Opposes the agonist; typically relaxes but can contract to decelerate/protect the joint. In climbing, pushing muscles are chronically under-trained antagonists.
Synergist	Assists the agonist and stabilises the joint. In elbow flexion, brachioradialis and brachialis are synergists to biceps.

Muscle Role	Definition & Climbing Example
Fixator (Stabiliser)	Stabilises the origin/joint so the agonist functions effectively. Rotator cuff fixates the shoulder during a pull-up.

ANTAGONIST MUSCLES — A CRITICAL COACHING CONCEPT

Climbing is a pulling-dominant sport. Climbers typically overdevelop pulling muscles (lats, biceps, finger flexors) relative to pushing muscles (chest, triceps, shoulder external rotators, finger extensors). This imbalance is a leading cause of shoulder, elbow, and finger injuries. A well designed programme ALWAYS includes dedicated antagonist training.

Section 6: Types of Muscle Contraction

Understanding how muscles contract is fundamental to coaching exercise technique safely and effectively. Muscles produce force three distinct ways, depending on whether they shorten, lengthen, or stay the same length under load:

6.1 Concentric Contraction

Occurs when a muscle shortens under load to overcome resistance — the most commonly recognised type.

- **Example:** Biceps shortens concentrically pulling the body up in a pull-up.
- **Climbing application:** Any pulling/gripping movement overcoming resistance — pulling through a move, initiating a deadpoint.

6.2 Eccentric Contraction

Occurs when a muscle lengthens under load — resisting rather than producing movement, typically against gravity.

- **Example:** Biceps contracts eccentrically controlling the descent phase of a pull-up.
- **Climbing application:** Lowering off, downclimbing, landing from a jump. Eccentric loading is the most powerful stimulus for tendon adaptation.
- **Important:** Eccentric contractions generate more force and cause greater muscle damage — the primary cause of DOMS. Heavy eccentric work needs longer recovery.

6.3 Isometric Contraction

Occurs when muscles develop tension with no movement — muscle length stays constant. The dominant, underappreciated contraction type in climbing.

- **Example:** Holding a lock-off at 90° elbow flexion — force generated, no shortening or lengthening.
- **Climbing application:** Holding a position between moves, sustaining a heel hook, gripping a hold. Most forearm work in climbing is isometric.
- **The occlusion effect:** Sustained isometric contraction restricts blood flow to the forearm flexors — the primary mechanism behind forearm pump, and why ARC training is effective for endurance.

CLIMBING IS LARGELY AN ISOMETRIC SPORT

Unlike running or cycling (predominantly concentric/eccentric), climbing demands sustained isometric contractions in the forearms and fingers for much of the ascent — making local muscular endurance a primary physical quality for climbing, distinct from cardiovascular endurance.

Section 7: Muscle Fibre Types

There are three types of skeletal muscle fibre, differing in contraction speed, force production, fatigue resistance, and energy system reliance. Understanding fibre types helps coaches prescribe climbing-specific training.

7.1 The Three Fibre Types

Fibre Type	Characteristics & Climbing Relevance
Type I — Slow Twitch	Contract slowly, resist fatigue over long periods. Rich in mitochondria/capillaries ('red fibres'). Aerobic. In climbing: ARC training, long easy routes, recovery.
Type IIa — Fast Twitch (Intermediate)	Fast contraction; uses both aerobic and anaerobic energy. Moderate fatigue resistance. In climbing: sustained hard bouldering, 4x4s, crux sequences.
Type IIb — Fast Twitch (Power)	Extremely rapid, very high force, fatigues quickly. Anaerobic only. In climbing: max moves, campus board, heavy hangboard, explosive dynos.

7.2 Muscle Fibre Characteristics Comparison

Characteristic	Type I	Type IIa	Type IIb
Contraction speed	Slow (90–140 ms)	Fast (50–100 ms)	Very fast (40–90 ms)
Force production	Low	High	Very high
Fatigue resistance	High	Intermediate	Low
Energy system	Aerobic (oxidative)	Both aerobic + anaerobic	Anaerobic only
Capillary density	High	Intermediate	Low
Mitochondria	High	Medium	Low
Glycolytic capacity	Low	High	High
Climbing use	Long routes, ARC, recovery	4x4s, power endurance	Max moves, campus, single hard efforts

7.3 What Determines Fibre Type?

1. **Genetics** — genetically predisposed to a certain fibre type percentage. Average ~60% fast twitch / 40% slow twitch, with considerable variation.
2. **Hormone levels** — fluctuate with age and training type. Heavy strength training elevates anabolic hormones supporting hypertrophy.
3. **Training undertaken** — fibres adapt behaviourally to demands placed on them, even if true type conversion is debated.

COACHING IMPLICATION

A climber genetically gifted with more fast-twitch fibres may excel at bouldering but struggle with endurance routes. Coaches should address individual weaknesses rather than assuming a fixed 'type' — slow-twitch dominant athletes can build power endurance, and fast-twitch dominant athletes can develop route endurance with appropriate programming.

Section 8: How Muscles Contract — Physiology

Understanding contraction physiology helps coaches explain what happens during training, why fatigue occurs, and why certain methods work.

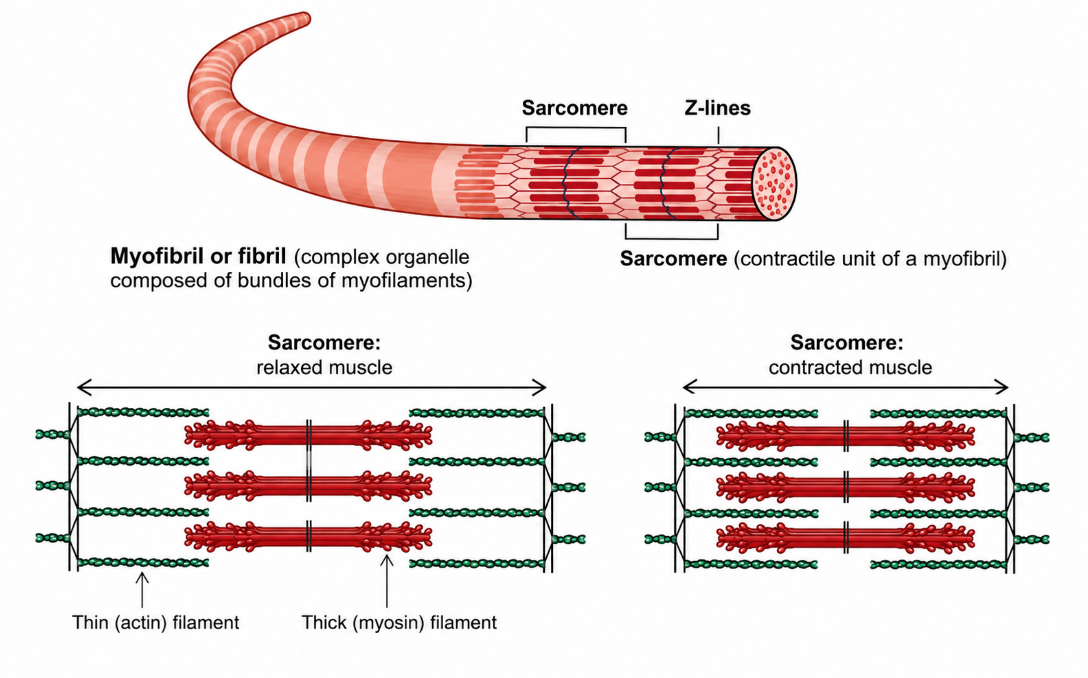
8.1 The Sliding Filament Theory

Within each sarcomere, actin and myosin filaments interact to produce contraction, in four stages:

1. **Muscle activation** — a motor neuron impulse triggers calcium release from the sarcoplasmic reticulum.
2. **Muscle contraction** — calcium binds troponin, exposing myosin binding sites; myosin heads pull actin toward the sarcomere centre (the 'power stroke'), consuming ATP.
3. **Recharging** — ATP is re-synthesised, allowing myosin to detach and reset. More cross-bridges formed equals greater force.
4. **Relaxation** — calcium is pumped back, myosin detaches, muscle returns to resting length. Also occurs when ATP is exhausted (failure).

THE THREE REQUIREMENTS FOR CONTRACTION

For a skeletal muscle to contract: (1) a neural stimulus, (2) calcium present, (3) ATP available. In climbing: muscle failure mid-route is primarily ATP depletion, not neural fatigue or calcium shortage.



Sarcomere structure — relaxed versus contracted muscle

8.2 Motor Units and Force Grading

A motor unit is a motor neuron and all fibres it innervates — firing contracts all fibres simultaneously ('all-or-none'). Force is graded via:

- **Motor unit recruitment** — small units recruited first (fine control), progressively larger ones as force demand increases (the Size Principle).
- **Rate coding** — increasing impulse frequency increases force ('summation'), reaching maximum continuous contraction ('tetanus') at high frequency.

Fingers use many small motor units for fine control; glutes use fewer, larger units for power — which is why finger technique and precise footwork take years to develop.

8.3 Muscle Spindles and the Stretch Reflex

Muscle spindles detect changes in muscle length and speed. Rapid lengthening triggers the stretch reflex — a spinal cord reflex (bypassing the brain for speed) causing the stretched muscle to contract. The faster the stretch, the stronger the reflex — the basis of plyometric training.

8.4 The Stretch-Shortening Cycle

The SSC occurs when a muscle is rapidly stretched immediately before contracting concentrically, producing more force through three factors:

1. **Loading the muscle spindle** — pre-loads neural drive before the concentric phase.
2. **Elastic energy storage** — like a rubber band, stored elastic energy releases as additional force.
3. **Optimal contraction length** — muscles generate greatest force at ~1.2x resting length; pre-stretching achieves this.

In climbing: the SSC is visible in dynamic moves. The brief dip before a dyno is not wasted — it exploits the SSC to generate more upward force.

SSC COACHING APPLICATION

When coaching dynamic moves, encourage a brief, controlled dip (countermovement) before the explosive phase — preloading spindles and storing elastic energy for more power than a static initiation.

Section 9: Major Muscles of the Body

The human body contains over 600 muscles. Understanding the primary movers (agonists) across each major joint is essential.

MUSCLES REQUIRED FOR THIS COURSE

This course doesn't require you to know all the muscles, their origin, insertion and innervations. The Muscle Finder app (corelearning.co.za/resources/muscle-finder.html) will help you find muscles relevant to specific climbing movements. Make yourself familiar with it and complete the quiz.

9.1 Shoulder & Chest

Muscle	Origin	Insertion	Action
Pectoralis major	Clavicle, sternum, ribs 1-6	Humerus (intertubercular groove)	Flexion, adduction, medial rotation

Muscle	Origin	Insertion	Action
Deltoid	Clavicle, acromion, scapular spine	Deltoid tuberosity of humerus	Abduction (mid); flexion/extension (ant/post)
Rotator Cuff (4)	Various scapular surfaces	Greater/lesser tubercle of humerus	Stabilise glenohumeral joint; rotation
Trapezius	Occipital bone, C7-T12	Clavicle, acromion, scapular spine	Elevation, retraction, rotation of scapula
Serratus anterior	Ribs 1-8	Medial border of scapula	Protracts/rotates scapula; holds against thorax

9.2 Upper Arm

Muscle	Origin	Insertion	Action
Biceps brachii	Coracoid process & supraglenoid tubercle	Radial tuberosity	Elbow flexion; forearm supination
Brachialis	Anterior humerus (distal half)	Ulnar tuberosity	Elbow flexion (primary)
Triceps brachii	Infraglenoid tubercle & posterior humerus	Olecranon of ulna	Elbow extension

9.3 Core & Trunk

Muscle	Origin	Insertion	Action
Rectus abdominis	Pubic crest & symphysis	Xiphoid process, ribs 5-7	Trunk flexion; compresses abdomen
External oblique	Ribs 5-12	Iliac crest, linea alba	Trunk rotation (contralateral), lateral flexion
Internal oblique	Thoracolumbar fascia, iliac crest	Ribs 9-12, linea alba	Trunk rotation (ipsilateral), lateral flexion
Transversus abdominis	Thoracolumbar fascia, iliac crest, ribs 7-12	Linea alba, pubic crest	Compresses abdomen; spinal stabilisation
Erector spinae (group)	Sacrum, iliac crest, ribs, vertebrae	Ribs, vertebrae, skull	Spinal extension and lateral flexion
Multifidus	Sacrum, ilium, vertebral spinous processes	Spinous processes 2-4 levels above	Deep spinal stabilisation; extension

9.4 Hip & Gluteal Region

Muscle	Origin	Insertion	Action
Gluteus maximus	Ilium, sacrum, coccyx	Gluteal tuberosity of femur; IT band	Hip extension, lateral rotation
Gluteus medius	External ilium	Greater trochanter of femur	Hip abduction; pelvic stabilisation
Gluteus minimus	External ilium (below medius)	Greater trochanter of femur	Hip abduction, medial rotation
Iliopsoas	Iliac fossa; T12-L5 vertebral bodies	Lesser trochanter of femur	Primary hip flexor
Hip adductors (group)	Pubis, ischium	Medial femur	Hip adduction; assist flexion/extension

Muscle	Origin	Insertion	Action
Piriformis	Anterior sacrum	Greater trochanter	Lateral rotation of hip; abduction when flexed

9.5 Thigh

Muscle	Origin	Insertion	Action
Quadriceps (4)	AIIS, ilium, linea aspera	Tibial tuberosity (patellar tendon)	Knee extension; rectus femoris flexes hip
Hamstrings (3)	Ischial tuberosity; biceps also from femur	Fibula head; medial/lateral tibia	Knee flexion; hip extension
Sartorius	ASIS	Medial tibia (pes anserinus)	Hip flexion, abduction, lateral rotation; knee flexion
TFL / IT band	ASIS, iliac crest	Lateral tibia (Gerdy's tubercle)	Hip abduction, medial rotation; knee stabilisation

9.6 Lower Leg & Foot

Muscle	Origin	Insertion	Action
Gastrocnemius	Medial/lateral femoral condyles	Calcaneus (Achilles tendon)	Plantarflexion; knee flexion
Soleus	Posterior fibula and tibia	Calcaneus (Achilles tendon)	Plantarflexion (primary)
Tibialis anterior	Lateral tibia	1st cuneiform, 1st metatarsal	Dorsiflexion; foot inversion
Peroneals (group)	Fibula	Metatarsals, calcaneus	Plantarflexion; foot eversion

9.7 Spinal Nerve Innervation

Each skeletal muscle receives its motor signal via specific spinal nerve roots — useful for correlating weakness with spinal injury level:

Nerve Root	Muscles Innervated
C3–C5	Diaphragm (phrenic nerve); trapezius (accessory nerve)
C5–C6	Deltoid, biceps brachii, brachialis, brachioradialis
C6–C7	Wrist extensors (radial nerve), pectoralis major
C7–C8	Triceps brachii, wrist flexors, finger extensors
C8–T1	Intrinsic hand muscles
T1–T12	Intercostals, erector spinae, abdominal muscles (by level)
L1–L3	Hip flexors (iliopsoas), hip adductors
L2–L4	Quadriceps femoris (femoral nerve)
L4–L5	Tibialis anterior, gluteus medius, hamstrings (in part)
L5–S2	Gastrocnemius, soleus, gluteus maximus, peroneals

9.8 The Kinetic Chain Concept

The kinetic chain is the interconnected system of bones, joints, and muscles working together. In Closed Kinetic Chain (CKC) movement the distal segment is fixed (e.g. a squat); in Open Kinetic Chain (OKC) movement it moves freely (e.g. a leg extension).

Term	Definition / Description
Closed Kinetic Chain (CKC)	Distal segment fixed. E.g. squat, deadlift, press-up, lunge. Multiple joints move together; more functional/sport-specific; greater joint stability.
Open Kinetic Chain (OKC)	Distal segment free. E.g. leg extension, bicep curl. Isolates single joint/muscle; useful in rehab; less compressive joint load.

PROGRAMMING IMPLICATION

A well-balanced programme should include both CKC and OKC exercises. CKC movements typically form the foundation of strength and conditioning due to functional transfer, while OKC exercises are valuable for targeted muscle development and rehabilitation.

End of Lesson 02. Return to corelearning.co.za to complete the Lesson 02 quiz (80% pass mark) and unlock the next lesson.